Kinam Kim

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Research Interest

My research is on **robot foundation models**: **Vision-Language-Action (VLA) models** [U1, C4] and **world models** built on video generative priors [C3, C5-C6, P1]. I work on what it takes to make them run on real robots: latent reasoning that stays inside the policy [U1], reinforcement learning for high-dimensional control and zero-shot sim-to-real transfer [C1, C2], and dexterous manipulation data generated at scale without teleoperation [O1].

Keywords: Vision-Language-Action (VLA) Models, World Models and Video Generation, Reinforcement Learning and Sim-to-Real, Dexterous Manipulation

Honors and Awards

- **Outstanding Paper Award**, Robotics: Science and Systems (RSS) 2026, for *FlashSAC* [C2]

Education

KAIST, Graduate School of AI — Daejeon, South Korea
M.S/Ph.D Integrated (Advisor: Prof. Jaegul Choo) — *Feb. 2025 - Present*
- GPA: 4.0 / 4.3

KAIST, School of Computing — Daejeon, South Korea
B.S. in Computer Science — *Mar. 2019 - Feb. 2024*
- GPA: 3.9 / 4.3

Work Experience

NVIDIA
Research Intern — *Jul. 2026 - Present*
- Latent-reasoning VLA with directly supervised reasoning states, under review [U1]

Microsoft Research Asia — Tokyo, Japan
Research Intern — *Feb. 2026 - Jul. 2026*
- Object-centric residual RL for VLA policies, zero-shot sim-to-real transfer (CoRL 2026) [C1]

KAIST DAVIAN Lab — Daejeon, South Korea
Undergraduate Research Intern — *Aug. 2023 - Mar. 2025*

Skills

- **Languages and frameworks:** Python, PyTorch, multi-GPU and multi-node distributed training
- **Robot learning:** VLA models (GR00T, π_0, SmolVLA), flow-matching policies, residual and off-policy RL

- **Simulation and real robots:** MuJoCo, RoboCasa, DexMimicGen, SO-101, sim-to-real transfer
- **Generative models:** video diffusion fine-tuning (Wan2.1, CogVideoX, Mochi), LoRA
- **Open source:** *Spatiotemporal Skip Guidance* [C6] in LTX-Video and Hugging Face Diffusers

Academic Service

- Reviewer: ICLR 2027, NeurIPS 2026, CoRL 2026

Publications

Under Review

[U1] **GistAct: Acting on Latent Answers in Vision-Language-Action Models**

Kinam Kim, Fred Yang, Min-Hung Chen, Zhiding Yu, Yu-Chiang Frank Wang, Chi-Pin Huang

Under review

Work done during an internship at NVIDIA

TL;DR: GistAct supervises a VLA's latent reasoning directly instead of distilling it from a text chain-of-thought teacher, ending its reasoning in a goal map over the scene. The action expert never reads the instruction, only that map, and reaches 67.8% success on RoboSPA spatial-24 against 33.9% for the strongest baseline.

Conference Paper

[C1] **Object-Centric Residual RL for Zero-Shot Sim-to-Real VLA Enhancement**

Kinam Kim, Namiko Saito, Heecheol Kim, Katsushi Ikeuchi, Jaegul Choo, Yasuyuki Matsushita

Conference on Robot Learning (CoRL), 2026

Work done during an internship at Microsoft Research Asia

TL;DR: We propose an object-centric residual reinforcement learning framework that augments a base VLA policy with a residual policy trained in simulation. Our approach transfers zero-shot to the real world, substantially improving success rates on manipulation tasks such as pick-and-place and drawer closing.

[C2] **FlashSAC: Fast and Stable Off-Policy Reinforcement Learning for High-Dimensional Robot Control**

Donghu Kim*, Youngdo Lee*, Minho Park, **Kinam Kim**, I Made Aswin Nahendra, Takuma Seno, Sehee Min, Daniel Palenicek, Florian Vogt, Danica Kragic, Jan Peters, Jaegul Choo, Hojoon Lee

Robotics: Science and Systems (RSS), 2026

Outstanding Paper Award

TL;DR: FlashSAC cuts the gradient updates in Soft Actor-Critic and compensates with larger models and higher data throughput, bounding weight, feature and gradient norms so the critic stays stable at that scale. It beats PPO and strong off-policy baselines on over 60 tasks across 10 simulators, and trains sim-to-real humanoid locomotion in minutes rather than hours.

[C3] **EgoX: Egocentric Video Generation from a Single Exocentric Video**

Taewoong Kang*, **Kinam Kim***, Dohyeon Kim*, Minho Park, Junha Hyung, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

TL;DR: We generate egocentric video from a single exocentric video, turning third-person footage into a first-person view for VR/AR.

[C4] **ACG: Action Coherence Guidance for Flow-based VLA models**

Minho Park*, **Kinam Kim***, Junha Hyung, Hyojin Jang, Hoijeong Jin, Jooyeol Yun, Hojoon Lee, Jaegul Choo

IEEE International Conference on Robotics and Automation (**ICRA**), 2026

TL;DR: We introduce Action Coherence Guidance (ACG), a training-free test-time guidance method that reduces noise-induced jerks, pauses, and jitter in VLA model policies. We show that ACG improves action coherence and significantly boosts success rates in RoboCasa, DexMimicGen, and real-world SO-101 manipulation tasks.

[C5] **Temporal In-Context Fine-Tuning for Versatile Control of Video Diffusion Models**

Kinam Kim*, Junha Hyung*, Jaegul Choo

The Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025

TL;DR: Temporal In-Context Fine-Tuning (TIC-FT) inserts noisy buffer frames to adapt a pretrained video diffusion model for controllable generation, using as few as 10 to 30 samples.

[C6] **Spatiotemporal Skip Guidance for Enhanced Video Diffusion Sampling**

Junha Hyung*, **Kinam Kim***, Susung Hong, Min-Jung Kim, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

TL;DR: Spatiotemporal Skip Guidance (STG) is a training-free sampling method that raises video diffusion quality by simulating a weak model through selective layer skipping, without the diversity loss of CFG.

Ongoing Project

[O1] **Demonstration-Free Dexterous Manipulation Data at Scale**

Kinam Kim, Jaegul Choo

Ongoing research project, 2026

TL;DR: We generate bimanual dexterous manipulation data for training robot policies without a single human demonstration, on contact-rich tasks such as Rubik's cube turning, carabiner clipping and bolt fastening. A coding agent composes trajectory optimization, motion planning and scripted control into a task policy and revises its own code from simulation feedback, scaling where teleoperation and RL do not.

Preprints

[P1] **Cross-Frame Representation Alignment for Fine-Tuning Video Diffusion Models**

Sungwon Hwang*, Hyojin Jang*, **Kinam Kim**, Minho Park, Jaegul Choo

arXiv preprint, 2025

TL;DR: Cross-frame Representation Alignment (CREPA) regularizes video diffusion fine-tuning against neighboring frames, improving fidelity and cross-frame coherence under LoRA.

Reference

- **Jaegul Choo**, Ph.D. Advisor, KAIST AI • jchoo@kaist.ac.kr
- **Yu-Chiang Frank Wang**, Internship Manager; Research Director, NVIDIA • ycwang@ntu.edu.tw
- **Yasuyuki Matsushita**, Internship Manager; Senior Director, MSRA Tokyo • yasumat@ist.osaka-u.ac.jp